

WILMINGTON, NC, USA

WMO: 723020

Lat: 34.268N

Lon: 77.900W

Elev: 10

StdP: 101.20

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 13748

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-4.3	-2.5	-14.1	1.1	-0.1	-11.3	1.4	1.5	11.1	11.7	9.9	11.1	2.6	30	0.430

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	8.7	34.1	25.4	32.9	25.0	31.7	24.6	26.7	31.5	26.1	30.5	25.6	29.7	4.4	250

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
25.4	20.6	28.6	24.9	19.9	28.1	24.4	19.4	27.8	82.8	31.3	80.6	30.5	78.4	29.9	29.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 9.4	8.4	7.5	-7.9	36.4	2.0	1.2	-9.4	37.3	-10.5	38.0	-11.6	38.7	-13.0	39.5		
(5)			-9.3	27.9	1.9	0.5	-10.7	28.3	-11.8	28.6	-12.9	28.9	-14.3	29.3		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	17.9	8.0	9.5	12.8	17.6	21.6	25.6	27.1	26.5	23.8	18.7	13.2	9.9
	DBStd	7.92	5.75	5.11	4.96	4.23	3.53	2.68	2.06	2.07	2.63	4.10	4.73	5.31
	HDD10.0	296	109	66	29	2	0	0	0	0	0	1	21	68
	HDD18.3	1308	323	250	182	65	12	1	0	0	1	46	164	265
	CDD10.0	3183	47	52	116	228	361	467	532	510	416	272	117	66
	CDD18.3	1153	2	2	10	42	114	218	273	252	167	59	10	4
	CDH23.3	9903	4	9	62	269	841	1974	2754	2344	1209	378	46	13
	CDH26.7	3341	0	0	7	50	238	711	1081	857	334	60	2	0
Wind	WSAvg	3.4	3.5	3.7	4.0	4.1	3.6	3.3	3.2	2.8	3.1	2.9	3.1	3.2
Precipitation	PrecAvg	1407	102	94	105	76	114	141	198	184	144	73	80	88
	PrecMax	1830	260	285	210	208	220	327	384	357	595	249	200	179
	PrecMin	959	17	26	24	12	24	23	42	42	18	10	12	15
	PrecStd	200	53	52	50	49	56	75	81	92	116	63	50	46
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	23.8	24.8	27.6	30.4	33.4	35.3	36.1	35.3	32.8	30.2	26.6	25.0
		MCWB	18.6	18.4	18.4	20.1	23.3	24.8	25.9	25.8	24.0	22.5	20.6	19.5
	2%	DB	21.4	22.4	25.1	28.0	31.0	33.5	34.2	33.3	31.2	28.2	24.7	22.8
		MCWB	17.5	16.8	17.8	19.7	22.7	24.7	25.7	25.6	23.7	21.8	19.2	18.8
	5%	DB	19.4	20.3	23.0	26.4	29.2	32.1	32.9	32.0	29.9	26.9	23.0	20.9
		MCWB	16.3	15.8	16.8	18.8	21.9	24.3	25.4	25.1	23.4	21.2	18.5	18.0
	10%	DB	17.5	18.3	21.0	24.7	27.6	30.6	31.6	30.8	28.7	25.6	21.4	19.0
		MCWB	14.7	14.5	16.1	18.0	21.1	23.9	25.0	24.7	22.9	20.8	17.8	16.6
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	20.0	20.0	20.8	22.9	24.9	26.8	27.5	27.4	25.9	25.0	22.4	21.6
		MCDB	22.3	22.9	24.3	27.2	30.3	32.6	33.2	32.5	29.3	27.3	24.3	23.4
	2%	WB	18.5	18.7	19.5	21.5	23.9	25.8	26.7	26.6	25.2	23.9	21.1	19.7
		MCDB	20.5	21.3	22.6	25.5	28.7	31.0	31.7	31.1	28.4	26.4	23.2	21.5
	5%	WB	16.9	17.1	18.4	20.5	23.0	25.2	26.1	26.0	24.7	22.9	19.8	18.4
		MCDB	19.0	19.2	21.6	24.2	27.0	29.9	30.7	30.0	27.8	25.5	22.0	20.4
	10%	WB	15.2	15.2	17.2	19.5	22.3	24.6	25.6	25.5	24.2	21.8	18.4	16.8
		MCDB	17.3	17.9	20.3	23.0	26.2	28.8	29.9	29.3	27.2	24.5	20.8	18.9
Mean Daily Temperature Range	5% DB	MDBR	10.9	11.4	11.7	11.4	10.3	9.1	8.7	8.5	8.8	10.6	11.8	11.0
		MCDBR	12.7	12.9	13.0	12.2	11.1	10.3	9.7	9.5	9.8	10.9	12.4	11.8
		MCWBR	9.4	8.4	7.1	5.6	4.6	3.9	3.6	3.6	4.0	5.4	7.8	8.5
	5% WB	MCDBR	11.3	11.6	10.8	10.1	9.5	9.2	9.0	8.6	7.9	8.3	10.3	11.0
		MCWBR	9.4	9.1	7.4	5.8	4.4	3.9	3.8	3.5	3.8	4.8	7.8	8.9
Clear-Sky Solar Irradiance	taub	0.327	0.335	0.366	0.398	0.458	0.488	0.526	0.503	0.427	0.386	0.349	0.331	
	taud	2.517	2.489	2.427	2.324	2.210	2.171	2.079	2.141	2.350	2.435	2.492	2.535	
	Ebn at Noon	883	913	908	889	836	807	774	785	835	844	848	857	
	Edh at Noon	83	95	109	127	144	150	163	151	117	99	85	78	
All-Sky Solar Radiation	RadAvg	2.70	3.40	4.51	5.67	6.22	6.42	6.08	5.48	4.60	3.91	2.99	2.39	
	RadStd	0.17	0.32	0.37	0.37	0.43	0.37	0.37	0.33	0.34	0.42	0.22	0.18	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		+0.47	N/A	N/A	+0.44	N/A	N/A	N/A	N/A	+147	+105
(47) Regional (43 neighbors)		+0.47	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+167	+120

Nomenclature: See separate page